

Duncan McAlester

TS/SCI

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EXECUTIVE SUMMARY

Design systems leader and design technologist with 10+ years building and modernizing production-grade design systems in high-stakes, regulated environments. Extensive experience delivering DoD and government design systems under active security clearance, with firsthand understanding of classified workflows, constrained hardware, and windowless, low-luminance operating conditions.

Creator and long-term steward of the Astro UX Design System, grown from concept to a mission-critical platform adopted across 200+ programs. Recognized and respected thought leader in design system strategy within the DoD community, known for turning fragmented, compliance-heavy UI ecosystems into token-driven, themeable systems that stay aligned between Figma and code—reducing design-engineering drift and cutting development effort by 30–50%.

Fluent across the design-system stack: Figma (tokens, libraries, governance), semantic token architecture, CSS variables, Web Components, Storybook, and visual regression pipelines. Particularly strong in accessibility for low-vision users and constrained displays, translating real-world operational constraints into enforceable system standards.

A collaborative system owner embedded with frontend platform teams, proactive about adoption, documentation, and community health. Sets clear standards while giving teams practical paths forward—balancing rigor with pragmatism so systems scale and endure.

KEY QUALIFICATIONS

- **Token-Driven Systems:** Defined and implemented semantic design tokens (typography, color, status, density, interaction) enabling themeable, scalable component libraries across multiple platforms.
- **Accessibility & Compliance:** Established MIL-STD 1472 and MIL-STD 2525 compliance frameworks for high-stress, mission-critical interfaces optimizing contrast, legibility, focus states, and input targets for constrained environments.
- **Design-Engineering Bridge:** Directed migration to W3C Web Components for framework-agnostic adoption; maintained design-code parity through structured token pipelines and component specs.
- **Storybook & Documentation:** Introduced docs-as-code practices and component documentation that improved developer adoption and reduced design-engineering drift.
- **Community & Governance:** Founded an industry-wide community of practice; ran intake processes, office hours, and maintained transparent backlogs connecting designers, engineers, and government stakeholders.
- **Government/DoD Experience:** Active TS/SCI clearance. Proven success in military and highly regulated domains—including U.S. Space Force satellite operations and programs with Lockheed, Raytheon, and Northrop Grumman.

EXPERIENCE

DIRECTOR OF PRODUCT DEVELOPMENT, ROCKET — 2024-NOV 2025

Lead cross-functional team delivering design system innovations and products for mission critical satellites operations centers.

- Delivered front-end architecture for next-generation satellite ground system using Web Components, establishing component APIs, design tokens and micro-front end frameworks
- Established component governance processes balancing quality with team velocity, running lightweight intake and maintaining transparent backlog and virtuous cycle of project -> design system -> project lifecycle
- Partnered with engineering to implement theme switching and runtime theming, supporting light and dark mode across constrained government hardware
- Developed compliance frameworks embedding MIL-STD 1472/MIL-STD 2525 accessibility standards into core component specifications
- Created secure design tooling for classified environments
- Technical Advisor on the ISTO STD MOSA-IF-S-003 Vols 1 & 2

DESIGN SYSTEM PRODUCT DIRECTOR, ROCKET — 2019-2024

Owned end-to-end strategy, governance, and delivery of the Astro UX Design System—scaling adoption from 0 to 200K annual downloads across 40+ Defense/ Aerospace organizations in the US and Europe.

- Led two major component library rewrites including migration to W3C Web Components, keeping Figma and code in sync
- Created design token architecture enabling consistent theming at scale with semantic token structure supporting multiple themes
- Defined design tokens, component APIs, and documentation supporting multiple development platforms
- Shipped components used by 200+ projects across government and Fortune 500 organizations with 200K annual downloads
- Expanded Storybook coverage with visual regression testing across themes
- Developed prototype applications showcasing system integration—recognized by Dan Mall as exemplary design system storytelling
- Maintained visibility into in-flight work, answered design system questions, and proactively turned emerging patterns into documented components and tokens
- Hosted monthly workshops, office hours, and clear change communication including deprecation guidance
- Established bi-annual SSC/EGS UX/UI Conference in Colorado Springs
- Gained industry-wide recognition in the Design System Community for excellence

LEAD DESIGN ENGINEER, ROCKET — 2015-2017

Launched the SATCOM UX Design System and later the Astro UX Design System, commissioned by U.S. Space Force for mission-critical operations.

- Established the common look and feel for all of US Space Force satellite operations
- Architected modular component library in CSS (20+ components) cutting front-end development time by 30%
- Introduced systematic documentation practices improving adoption and maintainability

PRINCIPAL/FOUNDER, INDEPENDENT UX CONSULTANCY — 1998-2015

Built and operated successful UX consultancy for 17 years serving government, enterprise, and Fortune 500 Clients

- Designed and developed a custom pizza ordering system, reducing operational costs by \$20,000 annually across over 100 locations.
- Designed and engineered a calendaring system for Senator Ted Kennedy's office, centralizing data entry and cutting staff completion time by 50%, boosting operational efficiency.
- Created the front-end for a custom content management system used by more than 50 US Senate offices, demonstrating significant adoption and impact on legislative operations.
- Designed and implemented front-end solutions recognized with multiple Gold and Silver Mouse awards, highlighting excellence in Congressional websites.

“Duncan’s enthusiasm was contagious at Rocket — I always looked forward to getting his messages about newly released CSS/browser features or his “what if we tried this?” ideas for our design system/ component library. It’s this depth of coding knowledge in addition to his leadership skills sets him apart from others in his role. ”

– Kiley M.

<https://www.linkedin.com/in/dmcalester/details/recommendations/>

PUBLICATIONS

- Co-author: McAlester, Duncan. Capraro, Michelangelo. *Skip Intro: Usability and Interface Design for Macromedia Flash*, New Riders Press, 2002
- Co-author: McAlester, Duncan. Capraro, Michelangelo. *Macromedia Flash MX 2004 Magic*, New Riders Press, 2004
- ISTO STD MOSA-IF-S-003, Volume 1 & 2: MOSA User Experience/User Interface Standard

AWARDS

- Macromedia People's Choice Award for Branding & Corporate Identity
- Multiple Gold Mouse Awards for Best Congressional Web Site
- Multiple Silver Mouse Awards for Best Congressional Web Site

SELECT SPEAKING & TEACHING ENGAGEMENTS

- BayCHI, Bay Area Computer Human Interaction Group, XEROX PARC
- OCCHI, Orange County Computer Human Interaction Group
- FlashForward San Francisco, Flash and User Experience Design
- FlashForward New York, Flash and User Experience Design
- University of California at Irvine, Multimedia Instructor
- University of Nebraska at Omaha, Guest Lecturer
- USSF/SSC Enterprise Ground Services UX/UI Conference
- BMC3I Innovation Hub in Colorado Springs

EDUCATION

LAGUNA COLLEGE OF ART AND DESIGN — 1993 – 1998

Bachelor of Fine Arts

“*Working with Duncan is like working with a crystal ball. He is constantly thinking ahead, considering new developments in tech (especially AI), systems thinking, design systems and role of design in general.*”

– Andrew A.

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